

- 1] The components of the mass flow vector $\rho \vec{u}$ are $\rho u_x = 4x^2y$, $\rho u_y = xyz$, and $\rho u_z = yz^2$.
- Compute the net mass outflow through the closed surface formed by the planes, $x = 0$, $x = 1$, $y = 0$, $y = 1$, $z = 0$, $z = 1$.
 - Compute $\vec{\nabla} \cdot (\rho \vec{u})$ and integrate over the volume bounded by the surface defined in part a).
 - Explain why the results in a) and b) should, or should not, be equal.
- 2] For the following flows find the missing velocity component needed for the flow to satisfy the incompressible continuity equation
- $u_1 = x_1^2 + x_2^2 + a^2$, $v_2 = -x_1x_2 - x_2x_3 - x_1x_3$, $u_3 = ?$
 - $u_1 = \ln(x_2^2 + x_3^2)$, $u_2 = \sin(x_1^2 + x_3^2)$, $u_3 = ?$
 - $u_1 = ?$, $u_2 = \frac{x_2}{(x_1^2 + x_2^2 + x_3^2)^{3/2}}$, $u_3 = \frac{x_3}{(x_1^2 + x_2^2 + x_3^2)^{3/2}}$
- 3] Consider the velocity field $\mathbf{u} = (\alpha x/t, 0, 0)$ (in Cartesian), where α is a constant. The density of the fluid is spatially uniform, but time-dependent, i.e., ρ is a function of t .
- Determine the density field $\rho(t)$, given $\rho = \rho_0$ at $t = t_0$.
 - What are the temporal $(\partial \mathbf{u} / \partial t)$, advective $(\mathbf{u} \cdot (\nabla \mathbf{u}))$, and total $(D\mathbf{u} / Dt)$ accelerations for this flow field? What does $\alpha = 1$ imply?
- 4] Consider a two-dimensional flow with velocity components given by

$$\begin{aligned} v_1 &= c x_1 \\ v_2 &= -c x_2 \end{aligned}$$

Find expressions for the vorticity and the strain rate tensor, (i.e. calculate their components)

- 5] Consider a plane Couette flow of a viscous fluid confined between two flat plates a distance b apart. At steady state the velocity distribution is $u = Uy/b$ and $v = w = 0$, where the upper plate at $y = b$ is moving parallel to itself at speed U , and the lower plate is held stationary. Find the rates of linear strain, the rate of shear strain, and vorticity in the flow.